



American International University-Bangladesh (AIUB)

Department of Computer Science
Faculty of Science & Technology (FST)

PROJECT TITLE

Rent A House Platform - Revolutionizing the Real Estate Rental Experience in Bangladesh

A Software Engineering Project Submitted
By

Semester: Spring_23_24		Section: J	Group Number: 02	
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Software Requirements Specification (SRS) for "Rent A House" Platform

1. Introduction

1.1 Background:

The "Rent A House" platform aims to revolutionize the real estate rental experience in Bangladesh by providing a comprehensive web-based solution for property owners and tenants.

1.2 Objectives:

The primary objectives of the software include connecting property owners and tenants seamlessly, offering an efficient property listing, user-friendly search, secure booking system, and enhancing overall user experience.

1.3 Scope:

The software will cover property listing, search and filtering, booking system, user profiles, reviews, and a notification system.

2. Functional Requirements

2.1 Property Listing

1. Detailed Property Information Input for Owners:

1.1: Property Registration

- Owners must register an account on the platform using their email or social media accounts.
- Provide necessary personal and property ownership details for account verification.

1.2: Property Listing Creation

- Upon successful registration, owners can initiate the property listing process.
- Fields include address, size, number of rooms, description, amenities, and lease terms.

1.3: Media Upload

- Integrate a feature allowing owners to upload images and videos of the property.
- Implement drag-and-drop functionality or file selection for ease of use.

1.4: Verification and Submission

- Implement validation checks to ensure completeness and accuracy of entered information.
- Allow owners to review and edit their listing before final submission.
- Provide an option to save drafts for future editing.

Priority Level: High

Precondition: Property owner is logged into the system.

2. Image Upload for Showcasing the Property:

2.1: Media Upload Interface

- Provide a dedicated section within the property listing form for media upload.
- Allow owners to upload images directly from their device or select from existing files.

2.2: Image Management

- Enable owners to rearrange or delete uploaded images as needed.
- Implement image preview functionality to ensure correct selection and order.

2.3: Image Quality and Standards

- Provide guidelines for image size, resolution, and format to ensure optimal display quality.
- Automatically compress or resize images if necessary to maintain platform performance.

2.4: Multimedia Support

- Extend media upload capabilities to support video files for enhanced property showcasing.
- Ensure compatibility with popular video formats and streaming services.

Priority Level: Medium

Precondition: Property owner is logged into the system and has an active property listing.

3. Pricing and Availability Management:

3.1: Pricing Configuration

- Allow owners to set rental pricing based on desired terms (daily, weekly, monthly).
- Include fields for additional costs such as utilities, maintenance fees, and security deposits.

3.2: Availability Settings

- Implement a calendar interface for owners to specify property availability.
- Enable selection of available dates or date ranges for prospective tenants to view.

3.3: Real-Time Updates

- Ensure availability and pricing information is updated in real-time across the platform.
- Notify owners of any conflicting bookings or pricing changes to avoid discrepancies.

3.4: Dynamic Pricing (Optional)

- Offer optional dynamic pricing features based on demand, seasonality, or other factors.
- Provide tools for owners to adjust pricing dynamically or set automated pricing rules.

Priority Level: High

Precondition: Property owner is logged into the system.

2.2 Search and Filtering

4. Advanced Search Options:

4.1: User Interface Design

- Design a user-friendly search interface accessible from the platform's homepage.
- Include prominent search bar with advanced options link/button for detailed filtering.

4.2: Basic Search Criteria

- Offer basic search criteria such as location, property type, number of bedrooms, and price range.
- Ensure intuitive input fields with dropdown menus, sliders, or text boxes.

4.3: Advanced Filtering

- Implement advanced filtering options for more specific search requirements.
- Include filters for amenities and property features.

4.4: Search Results Presentation

- Display search results in a clear and organized manner, highlighting key property details.
- Provide sorting options based on relevance, price, or other relevant factors.

4.5: Saved Searches and Alerts

- Allow users to save their search criteria for future reference.
- Implement email or push notification alerts for new listings matching saved search preferences.

Priority Level: High

Precondition: Tenant is logged into the system.

5. Interactive Map Integration:

5.1: Map Interface Design

- Integrate a dynamic map interface into the platform's search and property listing pages.
- Utilize popular mapping services (e.g., Google Maps) for familiarity and functionality.

5.2: Map Controls

- Implement zoom and pan controls for users to explore the map interface.
- Include options to switch between map view, satellite view, and street view.

5.3: Filtering on Map

- Allow users to filter search results directly on the map interface.
- Enable users to draw custom search boundaries or radius filters on the map.

Priority Level: Medium

Precondition: Tenant is performing a property search.

2.3 Booking System

6. Secure Messaging System:

6.1: User Authentication

- Users must create accounts and authenticate their identities before accessing the messages.
- Implement secure authentication methods such as email verification and password encryption.

6.2: Messaging Interface

- Design a user-friendly messaging interface accessible from within the platform.
- Include features for composing messages, viewing message history, and managing conversations.

6.3: Encryption and Data Privacy

- Implement end-to-end encryption to ensure the security and privacy.
- Comply with data protection regulations to safeguard user information and communication data.

6.4: User Notifications

- Provide real-time notifications for new messages, message deliveries, and message read receipts.
- Allow users to customize notification preferences (e.g., email, push notifications).

6.5: Spam and Abuse Prevention

- Implement spam filters and moderation tools to prevent unsolicited messages.
- Allow users to report inappropriate messages and take actions against offending accounts.

Priority Level: High

Precondition: Users (property owner or tenant) are logged into the system.

7. Reservation Requests and Confirmations:

7.1: Property Booking Process

- Enable users to submit reservation requests directly from property listings.
- Include a "Book Now" or "Request Reservation" button for initiating the booking process.

7.2: Booking Details and Terms

- Display booking details including dates, rental pricing, and any additional terms and conditions.
- Require users to review and accept booking terms before submitting a reservation request.

7.3: Reservation Confirmation

- Upon owner approval, send confirmation messages to both the tenant and the owner.
- Include reservation details such as booking dates, rental amount, and payment instructions.

Priority Level: High

Precondition: Users (property owner or tenant) are logged into the system.

8. Calendar Integration:

8.1: Calendar Access

- Allow users to access and manage their rental property calendars from within the platform.
- Provide options for syncing with external calendar applications (e.g., Google Calendar, Outlook).

8.2: Reservation Synchronization

- Automatically update property calendars with reservation details upon booking confirmation.
- Ensure seamless synchronization to prevent double bookings and conflicts.

8.3: Availability Visibility

- Display property availability status on the calendar interface for both owners and tenants.
- Highlight reserved dates and indicate pending reservation requests.

8.4: Reminder Notifications

- Send reminders for upcoming reservations and important dates (e.g., check-in, check-out).
- Allow users to set custom reminders and notifications preferences.

8.5: Integration with Booking Process

- Enable users to initiate reservation requests directly from the calendar interface.
- Provide options for selecting available dates and submitting booking requests seamlessly.

Priority Level: Medium

Precondition: Property owner is logged into the system.

2.4 User Profiles and Reviews

9. User Profiles:

9.1: Registration and Authentication

- Users must register an account with valid credentials to access profile features.
- Implement secure authentication methods such as email verification and password encryption.

9.2: Profile Creation

- Provide a user-friendly interface for users to create and edit their profiles.
- Include fields for personal information, profile picture, and preferences.

9.3: Profile Management

- Enable users to manage their profiles, including updating personal information.
- Allow users to view their booking history, saved properties, and interactions with other users.

Priority Level: High

Precondition: Users (property owner or tenant) are logged into the system.

10. Review and Rating System:

10.1: Review Submission

- Allow users to submit reviews and ratings for properties and other users (e.g., landlords, tenants).
- Include text fields for detailed reviews and rating scales (e.g., stars, thumbs-up/down).

10.2: Moderation and Validation

- Implement moderation processes to review and approve submitted reviews
- Validate reviews to ensure authenticity and prevent fraudulent or spammy submissions.

10.3: Review Visibility

- Display reviews and ratings prominently on property listings and user profiles.
- Provide options to filter reviews based on criteria such as date, rating, and relevance.

10.4: Response Mechanism

- Allow property owners and users to respond to reviews and address any concerns or feedback.
- Display owner responses alongside user reviews to promote transparency and accountability.

Priority Level: High

Precondition: Users (property owner or tenant) have completed a transaction.

2.5 Notification System

11. Real-Time Notifications:

11.1: Notification Settings

- Provide users with options to customize their notification preferences.
- Allow users to choose notification channels and specify triggers for receiving notifications.

11.2: Personalization

- Personalize notifications based on user preferences and behavior.
- Provide options for users to tailor notification content and frequency according to their needs.

11.3: Notification Center

- Create a centralized notification center where users can view and manage their notifications.
- Allow users to mark notifications as read, archive them directly from the notification center.

Priority Level: High

Precondition: Users (property owner or tenant) have enabled notification settings.

3. Non-Functional Requirements

3.1 Usability

User Training: The system shall be designed to be user-friendly, requiring minimal training for both property owners and tenants.

Priority Level: High

3.2 Performance

Response Time: The system shall provide responses to user actions within an average of 3 seconds under normal operating conditions.

Priority Level: High

3.3. Security:

Data Encryption: All sensitive user data, including personal information and payment details, must be encrypted to protect against unauthorized access.

Secure Authentication: Implement secure authentication methods such as multi-factor authentication to prevent unauthorized access to user accounts.

Regular Security Audits: Conduct regular security audits and penetration testing to identify and address vulnerabilities in the system.

Priority Level: High

3.4. Reliability:

High Availability: The platform should be available and accessible to users with minimal downtime, aiming for at least 99.9% uptime.

Disaster Recovery: Implement measures for disaster recovery and data backup to ensure minimal data loss and quick system recovery in case of system failures or disasters.

Priority Level: Medium

4. Project Requirements

4.1 Tools

Testing Tools:

The system developers shall utilize Selenium tools for testing activities, with testing activities scheduled for completion by week 6.

Priority Level: Medium

5. References

[1] Iu, C.S. and Salikon, M.Z.M., 2022. Room Renting Management System. Applied Information Technology And Computer Science, 3(2), pp.1451-1469.

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[3] Jones, M., & Brown, P. (2020). "Digital Transformation in Real Estate: A Comprehensive Review." International Journal of Real Estate Studies, 14(1), pp. 51-69.